Intelligent Agent

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Outline

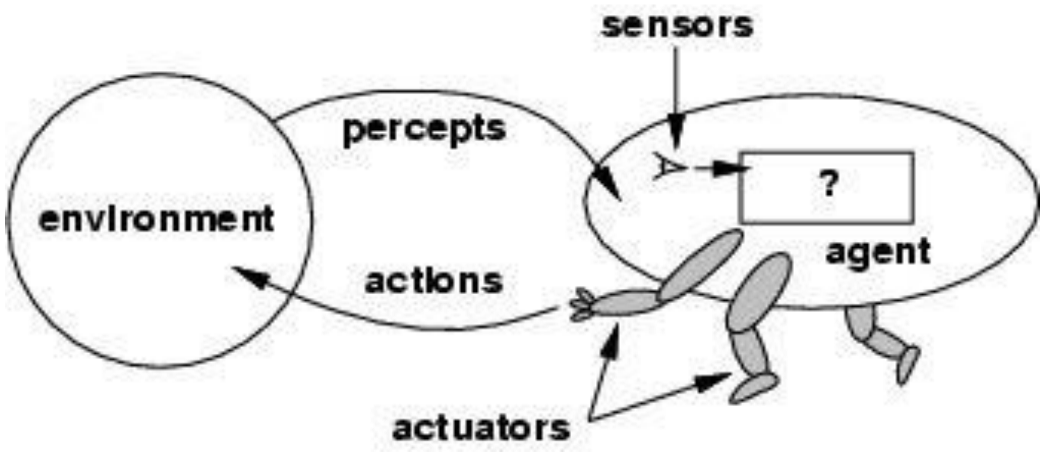
- Agents and environments
- PEAS (Performance measure, Environment, Actuators, Sensors)
- Rationality

Intelligent Agent

- Artificial Intelligence focuses on building systems that can act intelligently in real-world environments.
- Examples:
  - Google Maps
  - Chatbot
  - Autonomous vehicle

Intelligent Agents (continued)

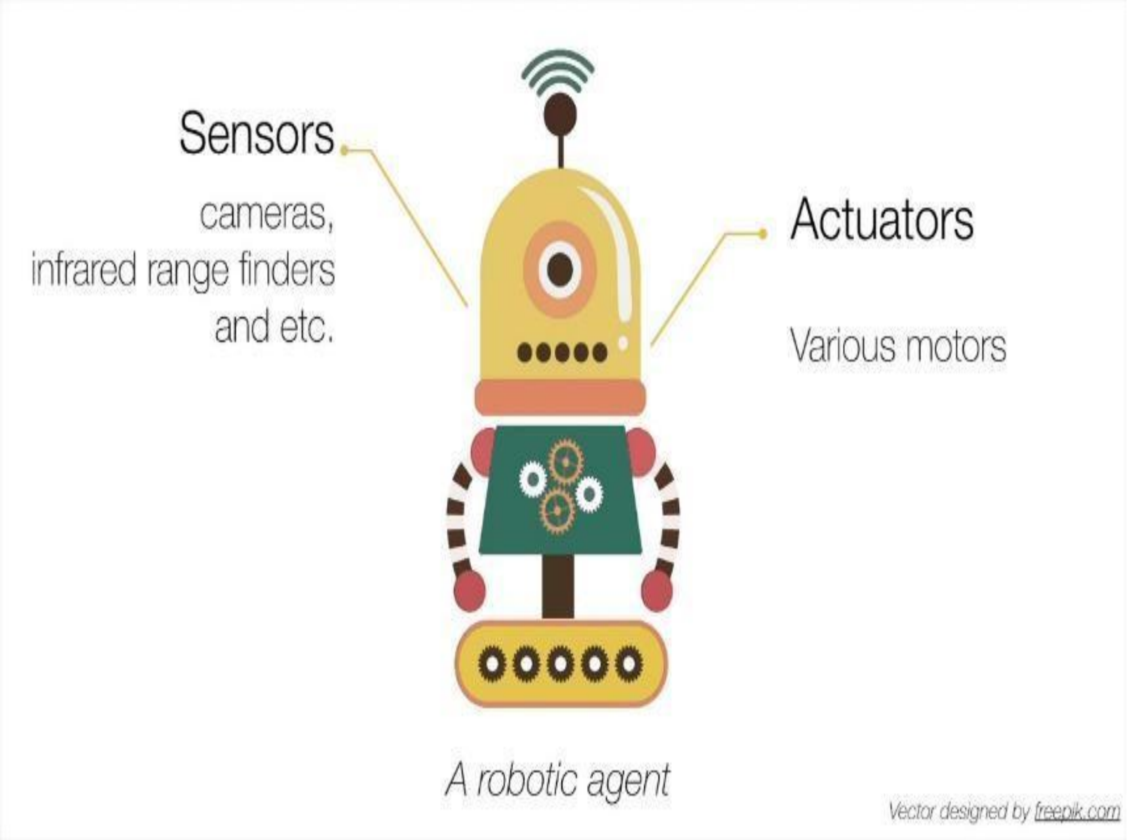
- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.



Intelligent Agents (continued)

- Sensor –
  - It is a device which detects the change in the environment and sends the information.
  - An agent observes it environment through sensors

- Actuators –
  - The component of machines that converts energy into motion.
  - The actuators are only responsible for moving and controlling a system.



Example - Agents

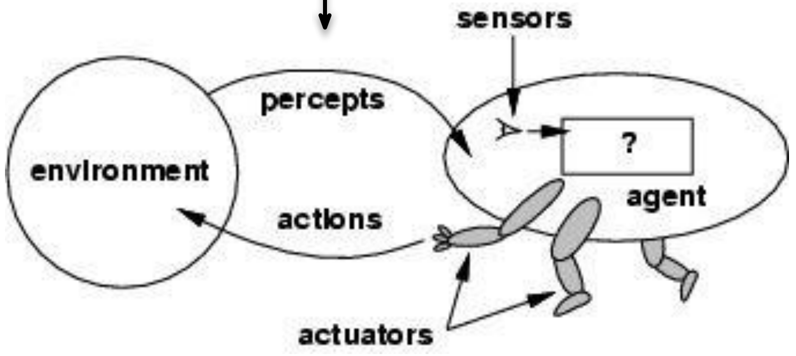
- Human agent:
  - eyes, ears, and other organs for sensors;
  - hands, legs, mouth, and other body parts for actuators
- Robotic agent:
  - cameras and infrared range finders for sensors;
  - various motors for actuators
- Software agent:
  - Keystrokes, file contents as sensors
  - Act on those inputs and display output called as actuators

Agent Terminology

- Percept
- Percept Sequence
- Behaviour of Agent
- Agent Function

Agents & Environment

Input as : see/ hear etc… (percept sequence)



- The agent function maps from percept histories to actions:[f: $\mathcal{P}^{*} \rightarrow \mathcal{A}$]
- The agent program runs on the physical architecture to produce f

agent = architecture + program

Agent Function vs Agent Program

Key Differences:

Aspect	Agent Function	Agent Program
Nature	Abstract and mathematical	Concrete and implemented
Purpose	Idealized mapping of inputs to actions	Practical execution of decisions
Flexibility	Considers all possible inputs	Limited by computational and design constraints
Real-World Use	Guides theoretical understanding	Operates in real environments

PEAS

- It is a systematic way to define the properties of an intelligent agent.
- It helps in designing and analysing agents by breaking down their interaction with the environment.

1. **P**erformance Measure
2. **E**nvironment
3. **A**ctuators
4. **S**ensors

Performance

- Behavior and performance of IAs in terms of agent function
    - Perception history (sequence) to Action Mapping
    - Ideal mapping: specifies which an agent should to take at any point in time.

- Performance measure: a subjective measure to characterize how successful an agent is (e.g., speed, power usage, accuracy, money, etc..)

Environment

- Refers to the **external world** in which the agent operates.
- Includes everything the agent can perceive and interact with.

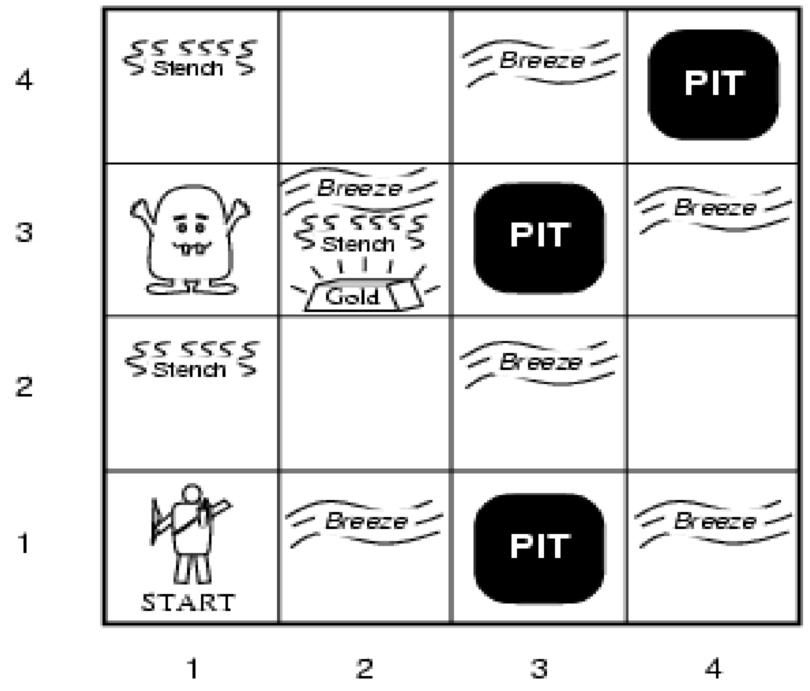
Sensors & Effectors

- An Agent perceives its environment through sensors
    - The complete set of inputs at a given time is called a **percept**
    - The current percept, or a sequence of percepts can influence the actions of an agent

- It can change the environment through actuators / effectors
    - An operation involving an actuator is called an action
    - Action can be grouped into action Sequence

PEAS – Wumpus World

- Performance measure
  - gold +1000, death -1000
  - -1 per step, -10 for using the arrow
- Environment
  - 4 X 4 grid of rooms
  - Agent always starts in square [1,1], facing to right
  - Squares adjacent to wumpus are smelly
  - Squares adjacent to pit are breezy
  - Glitter iff gold is in the same square
  - Shooting kills wumpus if you are facing it
  - Shooting uses up the only arrow
  - Grabbing picks up gold if in same square
  - Releasing drops the gold in same square



- Sensors:
  - Stench, Breeze, Glitter, Bump, Scream
- Actuators :
  - wheel/feet, arm

PEAS - taxi driver

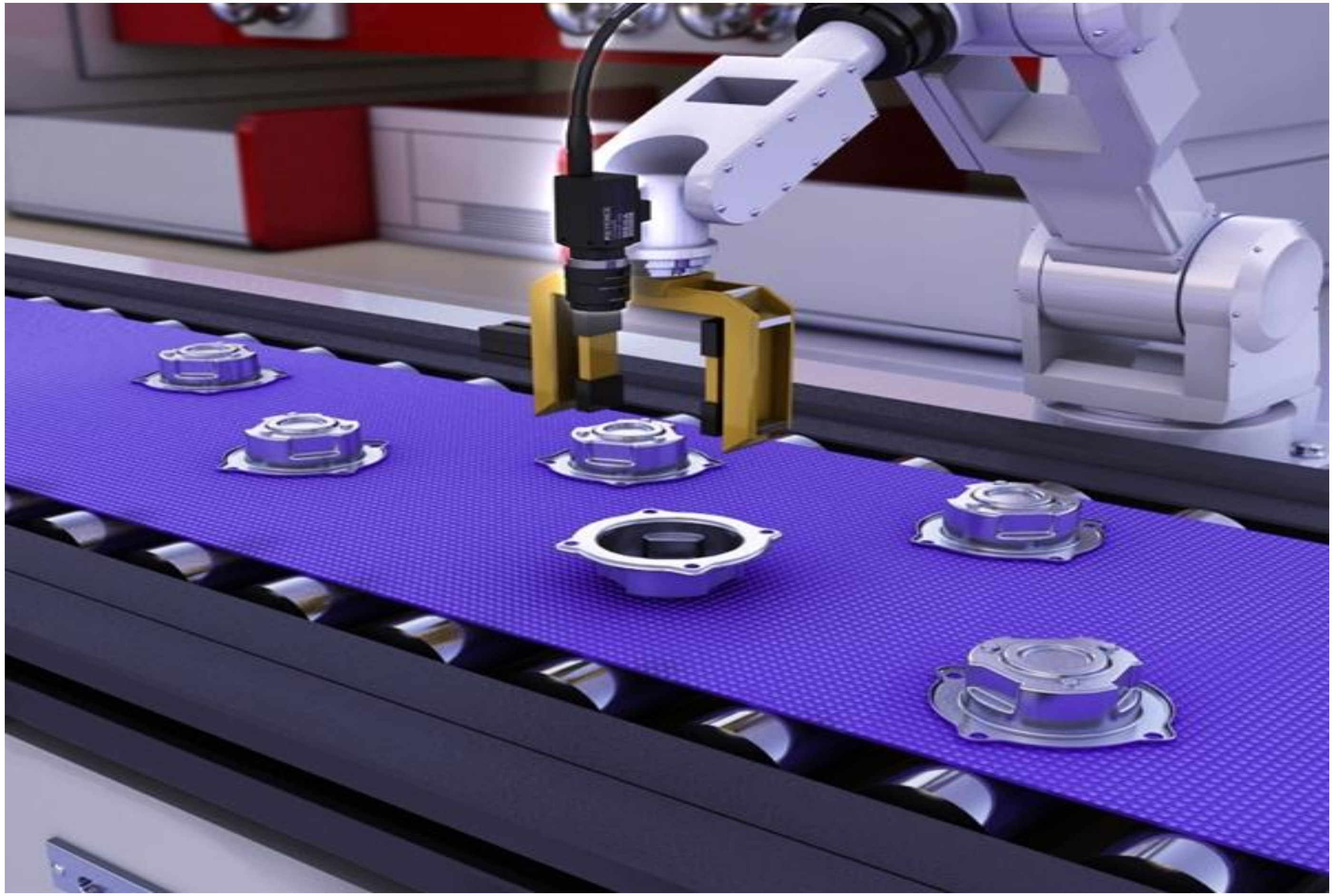
PEAS - taxi driver (continued)

- Agent Type
  - Taxi Driver
- Performance Measure
  - Safe, fast, legal, comfortable trip, maximize profits
- Environment
  - Roads, other traffic, pedestrians, customers
- Actuators
  - Steering, accelerator, brake, signal, horn, display
- Sensors
  - Cameras, sonar, speedometer, GPS, odometer, engine sensors, keyboard

Taxi – Driver Performance Measure

- Legal
  - Breaking signal ➔ -40 per signal break
  - Speeding ➔ -20 per speed limit violation
  - Wrong entry ➔ -30 for every wrong entry
  - Wrong parking => -20 for every wrong parking
  - Minor accident ➔ -70 for every minor accident
  - Major accident ➔ -100 for every major accident
- Safe
- Maximize profit
- Comfortable journey
  - Avoiding pothole ➔ +40 per pothole
  - Speeding at speedbreakers ➔ -30
  - Sudden break ➔ -30

Part-picking robot



Part-picking robot (continued)

- Agent – Part-picking robot
- Performance measure
  - Percentage of parts in correct bins
- Environment
  - Conveyor belt with parts, bins
- Actuators
  - Jointed arm and hand
- Sensors
  - Camera, joint angle sensors

Medical diagnosis

Medical diagnosis (continued)

- Agent Type
  - Medical diagnosis
- Performance Measure
  - Healthy patient, reduced costs
- Environment
  - Patient, hospital, staff
- Actuators
  - Display of questions, tests, diagnoses, treatments, referrals
- Sensors
  - Keyboard entry of symptoms, findings, patient’s answers

Online Shopping Recommendation System

- **Performance Measure:**
- **Environment:**
- **Actuators:**
- **Sensors:**

Smart Home Thermostat

- **Performance Measure:**
- **Environment:**
- **Actuators:**
- **Sensors:**

Drone Delivery System

- **Performance Measure:**
- **Environment:**
- **Actuators:**
- **Sensors:**

Functionalities of Rational Agent

- **Perceive:** Gather information from the environment through sensors.
- **Reason/Decide:** Process percepts to determine the best possible action.
- **Act:** Execute actions that influence the environment.
- **Learn:** Improve performance over time by adapting to new situations.
- **Achieve Goals:** Maximize success according to a performance measure (e.g., winning a game, cleaning a room efficiently).

Components of Rational Agent

- **Sensors:** Input mechanisms to perceive the environment.
- **Actuators:** Output mechanisms to act on the environment.
- **Knowledge Base:** Stores facts, rules, and learned information.
- **Decision-Making Module:** Chooses rational actions based on percepts and goals.
- **Learning Module:** Updates knowledge and strategies from experience.

Build Rational Agent

- **Define the Environment:** What the agent will operate in stochastic).
- **Specify Performance Measure:** What counts as success.
- **Design Architecture:** Combine sensors, actuators, and reasoning modules.
- **Implement Decision Logic:** Algorithms (rule-based, search, planning, learning).
- **Incorporate Learning:** Allow adaptation to improve over time.

Rational Action

- The action that maximizes the expected value of the performance measure, given the percept sequence so far.
- **Rational = Best?** Yes, but only relative to the agent's knowledge.
- **Rational = Optimal?** Yes, but only within the agent's abilities and constraints.
  - Rationality is about *doing the best you can with what you know and the resources you have*.
  - Optimality is the theoretical best possible action, which may not be achievable in practice.

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Intelligent Agent

Characteristics

- **Sense** → perceive the world.
- **Act** → influence the world.
- **Autonomous** → act independently.
- **Rational** → act logically to achieve goals.

Question ?